Problem Statement – 04

- **Problem Statement Title- AI-Powered Behavioral Anomaly Detection for Cybersecurity**
- **Theme - Cybersecurity**
- **PS Category - Software**
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- **Student ID - 23BAI10811**

AI-Powered Behavioral Anomaly Detection for Cybersecurity

Proposed Solution

- Hybrid AI system using **Isolation Forest + LSTM Autoencoder**.
- Detects behavioural anomalies in users, service accounts, and IoT devices.
- Classifies attacks and provides explainable alerts.

How it addresses the problem

- Learns normal behaviour instead of using fixed rules.
- Detects unknown and insider attacks in real time.
- Reduces false alerts with adaptive learning.

Innovation & Uniqueness

- Hybrid anomaly detection model.
- Handles **cold-start** and **concept drift**.
- Explainable AI with unified risk scoring.

TECHNICAL APPROACH

Flowchart

Access logs
↓
Feature Engineering
↓
Isolation Forest / LSTM Encoder
↓
Risk Score Fusion
↓
Attack Classification
↓
Explainable AI
↓
Security Dashboard

Tech Stack

- Python
- Pytorch
- pandas
- scikit learn
- NumPy
- JS
- CSS
- HTML

FEASIBILITY AND VIABILITY

Feasibility

- Open-source & cost-effective
- Scalable architecture
- Real-world deployment ready

Challenges

- Limited labelled data
- Cold-start detection
- Concept drift

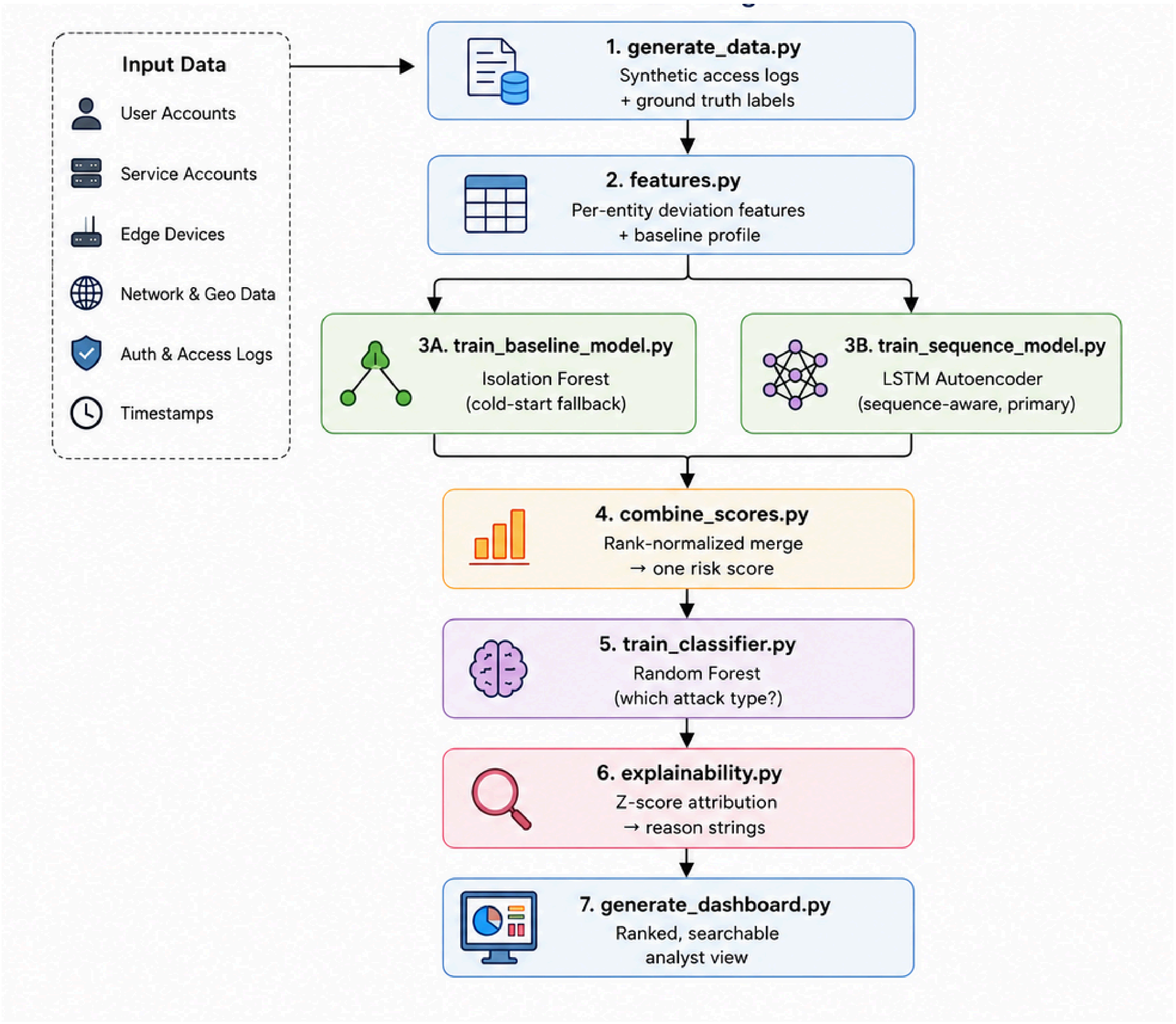
Mitigation Strategies

- Synthetic data generation
- Hybrid AI model
- Adaptive learning (EWMA)

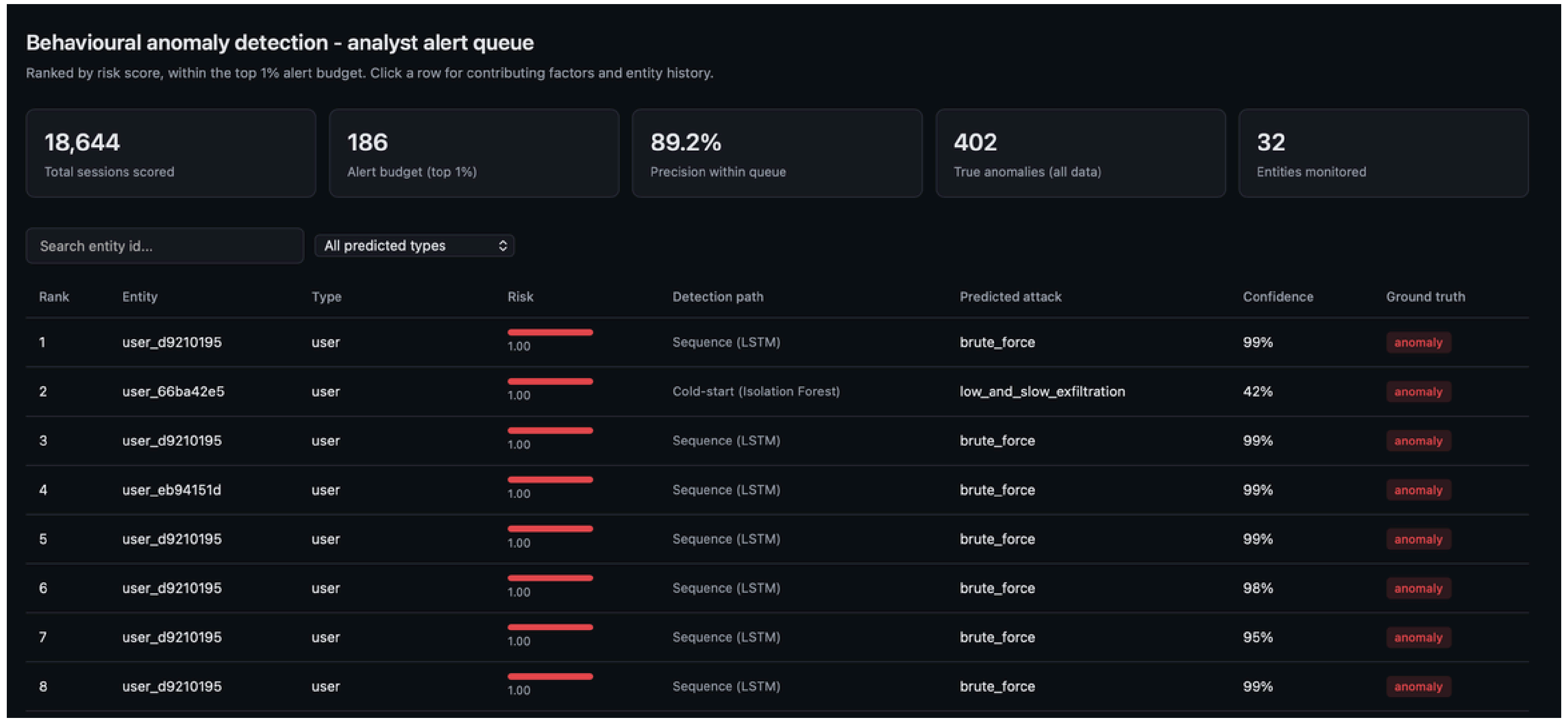
ARTIFACTS

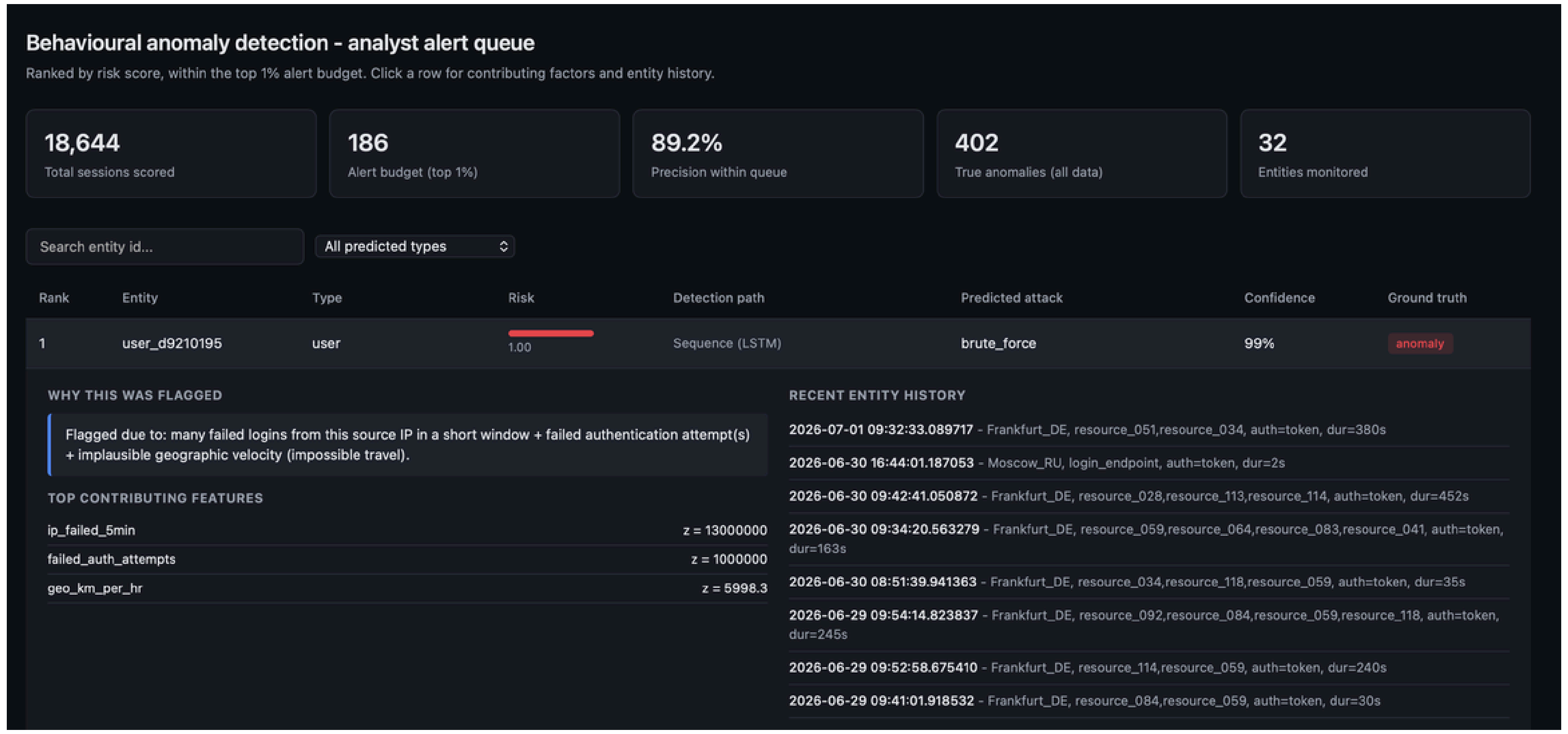
Github Link https://github.com/ayushikashyap1207/honeywell2

Architechture Diagram



Dashboard





RESEARCH AND REFERENCES

- **Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2008).** Isolation Forest. ICDM.
- **Hochreiter, S., & Schmidhuber, J. (1997).** Long Short-Term Memory (LSTM).
- **Sakurada, M., & Yairi, T. (2014).** Autoencoder-based Anomaly Detection.
- **scikit-learn Documentation**
- **PyTorch Documentation**